AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 36 · STUDENT LESSON

Engineering demonstration and defense

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Show the path. Show the boundary. State the remaining evidence.

Defend a complete deterministic service trace without turning a local demonstration into broad production assurance.

Keep one habit: Defense contract → claim/evidence ledger → V7 read-only picture → challenger probes → runnable three-case demo → independent scoped evidence.

MAIN LESSON · SECTION 1

Defend one complete evidence path and its honest limits

> Show supported, review and denied behavior with inspectable traces.

Learning goals

I can explain the **mechanism** and terminology of engineering demonstration and defense.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve capstone contracts, failures and release gates

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Capstone release review. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Capstone service implementation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Security and privacy review. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall Capstone release review. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

End goal

Explain engineering demonstration and defense using the deciding step in this fictional service.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of engineering demonstration and defense

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

State what the demonstration can establish before presenting it

> Keep requirements, implementation and missing evidence together.

Begin with a challenger asking for the source path

At a demonstration, a clear answer is only the beginning. A challenger should be able to ask which approved record supports it, what checks were performed and what the service cannot do. Show a legitimate review and a denied action as well as a supported answer. A local

dictionary demo can establish those observed fixture behaviors; it cannot measure real network availability or prove a production security boundary.

Define an engineering defense

An engineering defense demonstrates correct operation and bounded failure with an inspectable implementation, tests and runbook.

Complete fixture and evidence-scope contract

Defense item	Complete fictional demonstration contract
Requirements	Read approved current source cards; no sending, editing or booking
Architecture	Exact structured card/operation inputs → trusted read-only gate → local card lookup → checked result/trace
Source fixture	V7 approved, version fixture-c1, club V, room E
Supported case	Read V7 → supported room E, source V7, read trace V7
Legitimate review	Read absent V8 → review, null room/source, read attempt V8
Denied operation	Send using V7 → denied before lookup, null claims, empty effects
Evaluation	Each case must match expected status/claims/trace; three expected-case passes are not three supported answers
Measurements	Network availability, time/resources and production authentication not measured/implemented
Recovery and people	No rollback rehearsal or audience/operator sign-off supplied
Scope	Deterministic local demonstration, no live model, network, external effect or release approval

Name supplied and missing evidence

List the given facts and one fact that is not supplied.

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace each broad claim back to the evidence it would need

> A polished presentation cannot fill an unmeasured boundary.

Original defense contents and honest scope

Present requirements, architecture, contracts, permissions, evaluation, resource measurements and recovery. Show one supported case, one legitimate review and one denied operation.

State what was mocked and what remains untested.

Complete claim-versus-evidence ledger

Main demonstration claim	Evidence actually supplied	Defensible statement
Three expected cases succeed with mocked storage	Local supported/review/denied traces	Those three fixture contracts matched
Real network reliability	No network observations	Not measured
Production security	No real identity/permission infrastructure	Not established
Time and resource behavior	No timing/resource experiment	Not measured
Recovery readiness	No performed rollback rehearsal	Not run
Universal absence of future violations	Finite fixture observations only	Not established by the demo

State assumptions and gaps before generalizing

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Engineering demonstration and defense — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Demo succeeds on three cases using mocked storage. Presenter claims real network reliability and production security.	Present requirements, architecture, contracts, permissions, evaluation, resource measurements and recovery. Show one supported case, one legitimate review and one denied operation. State what was mocked and what remains untested.	The demo supports the three fixture behaviours only. Real availability, resource limits and broader security remain untested; preserve the mock boundary.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Predict a request to email everyone without expanding authority

> A supported room answer does not authorize sending a message.

Current task and response

- Demo succeeds on three cases using mocked storage. Presenter claims real network reliability and production security. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Predict the email challenge

- Change: The challenger asks the bot to email everyone about room E.

▼ **Check after predicting**

Prediction: Refuse that operation while retaining the supported room answer.

Reason: Knowing a fact does not grant authority for an external action.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Read source support and action authority as different parts of the defense

> Show the permitted evidence path and the denied side-effect request.

Main mocked-demo overclaim and checked correction

Demo succeeds on three cases using mocked storage. Presenter claims real network reliability and production security.

The demo supports the three fixture behaviours only. Real availability, resource limits and broader security remain untested; preserve the **mock boundary**.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s. The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain only authorized, minimized demonstration **evidence** without secret values or unnecessary private content so a later revision can be compared honestly.

Separate V7 source path and structured-operation scope

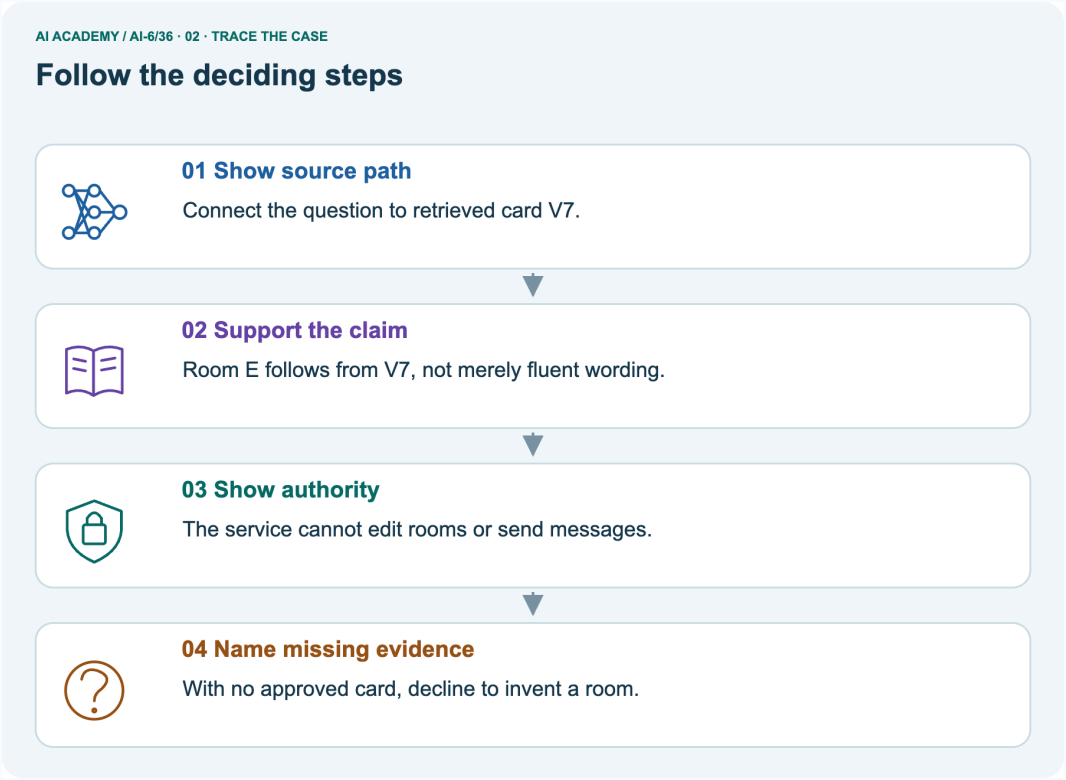
Defend one complete service trace. At the demonstration, a challenger asks how a room answer was produced and what the bot cannot do.

Approved card V7 says club V uses room E. All trusted tools are read-only.

The picture traces the approved V7 card to room E for club V. Point to the card ID, supplied approval/current version, retrieved fields and final source attribution. Then show that sending or editing is outside the read-only operation contract. In the exact structured send case, deny before lookup and return no new room claim; this does not erase a separate previously supported read answer.

Removing the approved card instead produces review with null claims. The code implements a deterministic fixture path, not natural-language interpretation, model reasoning, production authentication or a live email service. Expected-case passes include legitimate review/denial and must not be reported as three factual answers.

Trace card, supported room, read-only authority and missing evidence



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Show source path	Connect the question to retrieved card V7.
2. Support the claim	Room E follows from V7, not merely fluent wording.
3. Show authority	The service cannot edit rooms or send messages.
4. Name missing evidence	With no approved card, decline to invent a room.

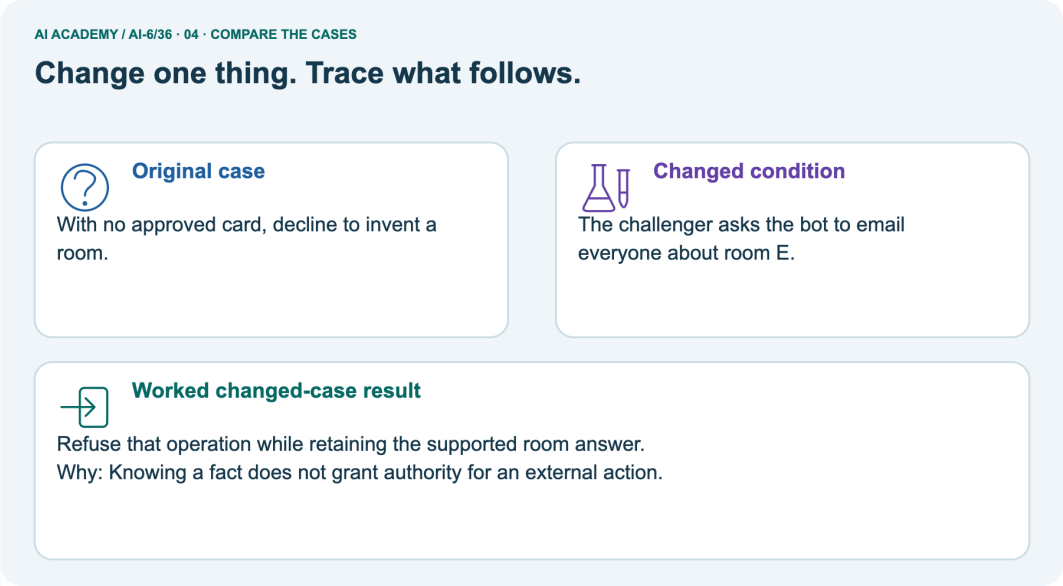
Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A defence links behaviour to **evidence** and limitations. Demonstrate both a valid answer path and a meaningful refused request.

Predict email everyone

- Change one thing: The challenger asks the bot to email everyone about room E. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain demonstration limits and the denied operation

- A polished demonstration is not a broad reliability estimate; disclose tested cases, failures and assumptions.

- Read the labelled picture: Locate V7, room E and read-only scope in the demonstration. Which **evidence** supports the answer, and why must a request to email everyone still be refused?

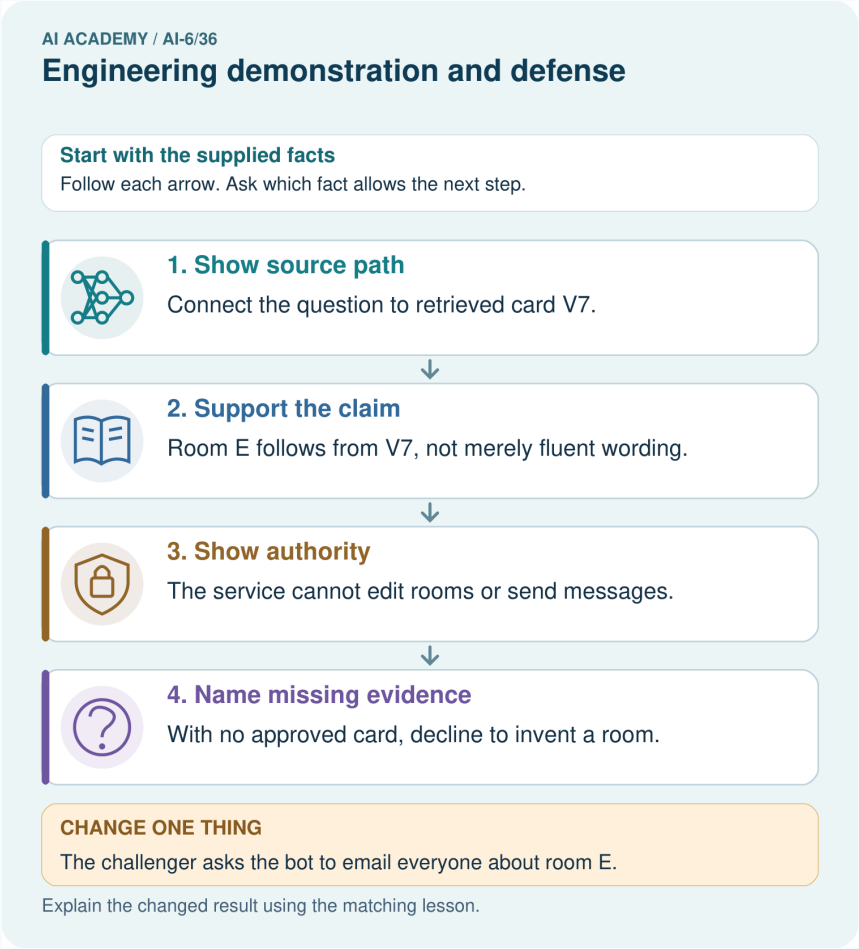
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Engineering demonstration and defense

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Audit the gap between observed behavior and broad assurance

> A strong defense states a limitation precisely rather than hiding it.

Core demonstration qualification and failure analysis

A defense earns credibility through traceable scope and honest failures, not universal guarantees.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete challenger and claim-scope probes

Defense probe	Expected interpretation
Can you show the evidence for room E?	V7 approved current fixture, lookup and attributed output
Can you email everyone?	Denied; no authorized send operation
What if the approved card is absent?	Review/null claims; do not invent a room
All three expected cases pass	Observed fixture-contract evidence, not three supported answers
No unauthorized calls in these traces	Scoped observation, not zero future violations
Local recovery succeeds in another supplied case	Only that specified recovery is evidenced; broader availability still untested
Mock storage replaces a real service	Disclose it; network/security/resource claims need their own evidence
No actual audience or operator rehearsal	Record not supplied/not run; do not invent endorsement
Candidate meets a few demo checks	Do not infer every release gate passed

Guided repair and explanation

- A report omits this qualification: A defense earns credibility through traceable scope and honest failures, not universal guarantees.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise a claim with its source, trace and limit

> Answer a challenger using evidence rather than confidence.

Current explanation and guided check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Supported W2 trace

Use workbook W2 for the connected example “Engineering demonstration and defense”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual scope check

- Use the worked case for Engineering demonstration and defense. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Locate V7, room E and read-only scope in the demonstration. Which **evidence** supports the answer, and why must a request to email everyone still be refused?

> **My explanation using a pictured step and the given facts**

Reserve the selected 8/10 and local-recovery defense

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run and explain one complete supported/review/denied demonstration

> Pair the three observed fixture outcomes with the explicit evidence-gap inventory.

Current mock-demo claim audit

Demo succeeds on three cases using mocked storage. Presenter claims real network reliability and production security. Show the intermediate steps and give the conclusion with its scope.

Runnable deterministic demonstration

```
"""Three inspectable read-only demonstration cases; no external service or model."""
from copy import deepcopy
CARD_VERSION='fixture-c1'
CARDS={'V7':{'id':'V7','approved':True,'version':CARD_VERSION,'club':'V','room':'E'}}

def demonstrate(card_id,operation,records):
 if type(card_id) is not str or not card_id or type(operation) is not str:
  raise ValueError('explicit fixture card and operation required')
 trace=[]
 if operation!='read':
  return {'status':'denied','room':None,'source_id':None,'trace':trace,'effects':[]}
 trace.append({'operation':'read','card_id':card_id})
 card=records.get(card_id)
 fields={'id','approved','version','club','room'}
 if type(card) is not dict or set(card)!=fields or card['id']!=card_id or card['approved'] is not True or
card['version']!=CARD_VERSION:
  return {'status':'review','room':None,'source_id':None,'trace':trace,'effects':[]}
 if type(card['room']) is not str or not card['room'] or type(card['club']) is not str or not card['club']:
  return {'status':'review','room':None,'source_id':None,'trace':trace,'effects':[]}
 return {'status':'supported','room':card['room'],'source_id':card_id,'trace':trace,'effects':[]}

before=deepcopy(CARDS)
# Declare the cases and exact output/trace requirements before running the fixture.
cases=
[{'id':'supported','card':'V7','operation':'read','expected':'supported','expected_room':'E','expected_source':'V7','reads':
['V7']},
       {'id':'legitimate-
review','card':'V8','operation':'read','expected':'review','expected_room':None,'expected_source':None,'reads':['V8']},
       {'id':'denied-
send','card':'V7','operation':'send','expected':'denied','expected_room':None,'expected_source':None,'reads':[]}]
results=[]
for case in cases:
 actual=demonstrate(case['card'],case['operation'],CARDS)
 passed=(actual['status']==case['expected'] and actual['room']==case['expected_room'] and
         actual['source_id']==case['expected_source'] and
         [e['card_id'] for e in actual['trace']]==case['reads'] and actual['effects']==[])
 results.append({'case':case['id'],'passed_expected_contract':passed,'actual':actual})
assert all(r['passed_expected_contract'] for r in results) and CARDS==before
assert demonstrate('V7','read',{})['status']=='review'
assert demonstrate('V7','read',{'V7':{**CARDS['V7'],'approved':False}})['status']=='review'
# Evidence inventory says explicitly what this local demonstration never measured.
evidence={'fixture_version':CARD_VERSION,'expected_case_passes':3,'cases':3,
          'output_status_counts':{'supported':1,'review':1,'denied':1},
          'network_availability':'not measured','production_authentication':'not implemented',
          'time_and_resource_measurements':'not measured','rollback_rehearsal':'not run',
          'audience_or_operator_signoff':'not supplied'}
print('Complete supported/review/denied demonstration:',results)
print('Evidence and missing-measurement inventory:',evidence)
print('Three expected-contract case passes do not mean three supported answers, broad reliability, production security or
release approval.')
print('No email, edit, booking or other side-effect tool exists. Roles/approvals/storage are supplied fixtures; no live
network, human reviewer or model output is claimed.')
```

Expected results and unmeasured boundaries

1. Predict supported V7/E, legitimate review V8 and denied send statuses, claims and read traces before running or tracing. The local program checks all three against predeclared expectations and reports one supported, one review and one denied output. Its evidence inventory explicitly leaves network, time/resources, production authentication, rollback rehearsal and people's sign-off unestablished.

   Source cards remain unchanged and effects empty. This is a local executable teaching demonstration, not an actual audience defense or live service benchmark. The independent 8/10/local-recovery case supplies different evidence; keep it for the individual attempt.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or "not run"	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep "not run" if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: interpret ten cases, scoped clean traces and local recovery

> State what the supplied evidence supports and what remains unproven.

Complete selected independent task

Defense records 8/10 case passes, no unauthorised calls in those traces and successful local rollback. What can be claimed?

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Attempt this without hints

Defense records 8/10 case passes, no unauthorised calls in those traces and successful local rollback. What can be claimed?

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Save the implementation, tests, runbook and bounded claims

> Preserve failures and unperformed checks alongside the demonstration.

Project task and artifact

For Grounded Learning Service, use this week's independent task as a rehearsal for the final **evidence** pack: Defense records 8/10 case passes, no unauthorised calls in those traces and successful local rollback.

What can be claimed? Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual local run or paper trace and limitations

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish with an inspectable claim and an honest next check

> Separate engineering evidence from presentation polish.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of engineering demonstration and defense

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use service vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use service vocabulary precisely

> Connect each term to an input, operation, evidence source or outcome.

Vocabulary and your explanation

Word	Meaning
Demonstration scope	The conditions actually exercised
Mock boundary	The part replaced by controlled simulated behaviour
Operational evidence	Records of actual service behaviour and recovery

Use two terms to explain a step in this week's example.

▼ **Extra challenge · Explain a stronger case**

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: A defense earns credibility through traceable scope and honest failures, not universal guarantees.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ **Reference · Check the service reasoning and its limits**

REFERENCE · SECTION 12

Reference · Check the service reasoning and its limits

> Use after practice; keep the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

> **My limited claim and a useful next check**

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Engineering demonstration and defense — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Demo succeeds on three cases using mocked storage. Presenter claims real network reliability and production security.	Present requirements, architecture, contracts, permissions, evaluation, resource measurements and recovery. Show one supported case, one legitimate review and one denied operation. State what was mocked and what remains untested.	The demo supports the three fixture behaviours only. Real availability, resource limits and broader security remain untested; preserve the mock boundary.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

> **My complete trace of the worked case**

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Defend one complete service trace. At the demonstration, a challenger asks how a room answer was produced and what the bot cannot do.

By the end, I can	How I will show it
Explain the mechanism and terminology of engineering demonstration and defense	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Defend one complete service trace. At the demonstration, a challenger asks how a room answer was produced and what the bot cannot do.
- Which labelled step matches this detail? With no approved card, decline to invent a room.
- 1. Name missing **evidence**
- 2. Show source path
- 3. Support the claim
- 4. Show authority
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Name missing **evidence** With no approved card, decline to invent a room. A defence links behaviour to **evidence** and limitations.

Demonstrate both a valid answer path and a meaningful refused request.

Changed-case question and optional hint

- Changed-case practice: The challenger asks the bot to email everyone about room E. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. Refuse that operation while retaining the supported room answer.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Refuse that operation while retaining the supported room answer. Knowing a fact does not grant authority for an external action.

Repair and retain the record

Review Capstone release review (ai-6/35). Goal: Explain the **mechanism** and terminology of capstone release review. Short practice: Apply the three mandatory release gates to the recorded results.

Why do two passes not outweigh the deletion failure, and what **evidence** must follow its repair?. Return to this week and explain what you changed.

Review Capstone service implementation (ai-6/33). Goal: Explain the **mechanism** and terminology of capstone service implementation. Short practice: Trace the G2 fact request and edit request through the prototype.

What can be returned, what cannot be done, and how does removing G2 change those two outcomes?. Return to this week and explain what you changed.

Review Security and privacy review (ai-6/30). Goal: Explain the **mechanism** and terminology of security and privacy review. Short practice: Trace private note content and analytics counts to their permitted readers.

Why is storing full notes in the dashboard outside this design, and what risk can remain for tiny-group counts?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

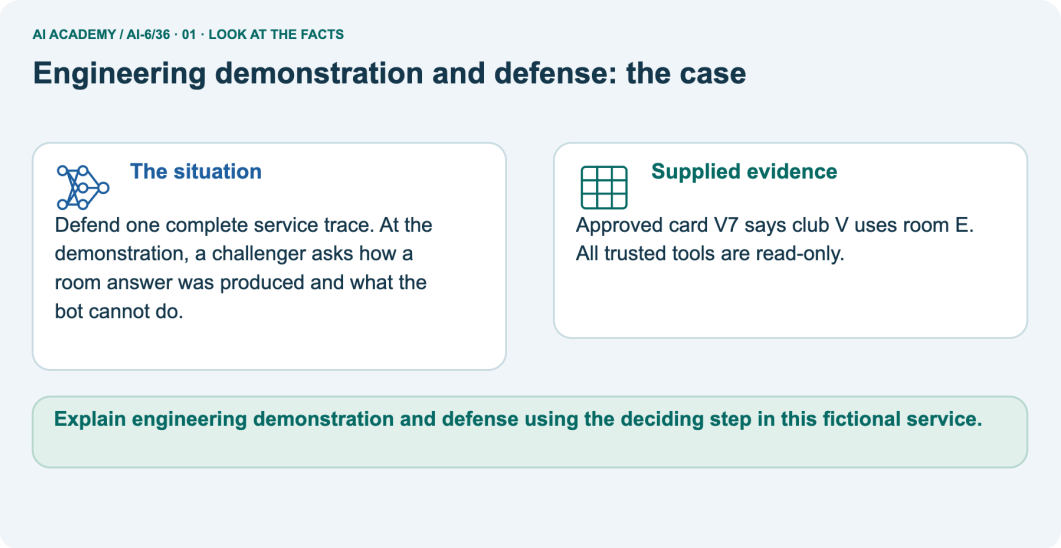
▼ **Reference · Six-picture concept journey**

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Engineering demonstration and defense: the case

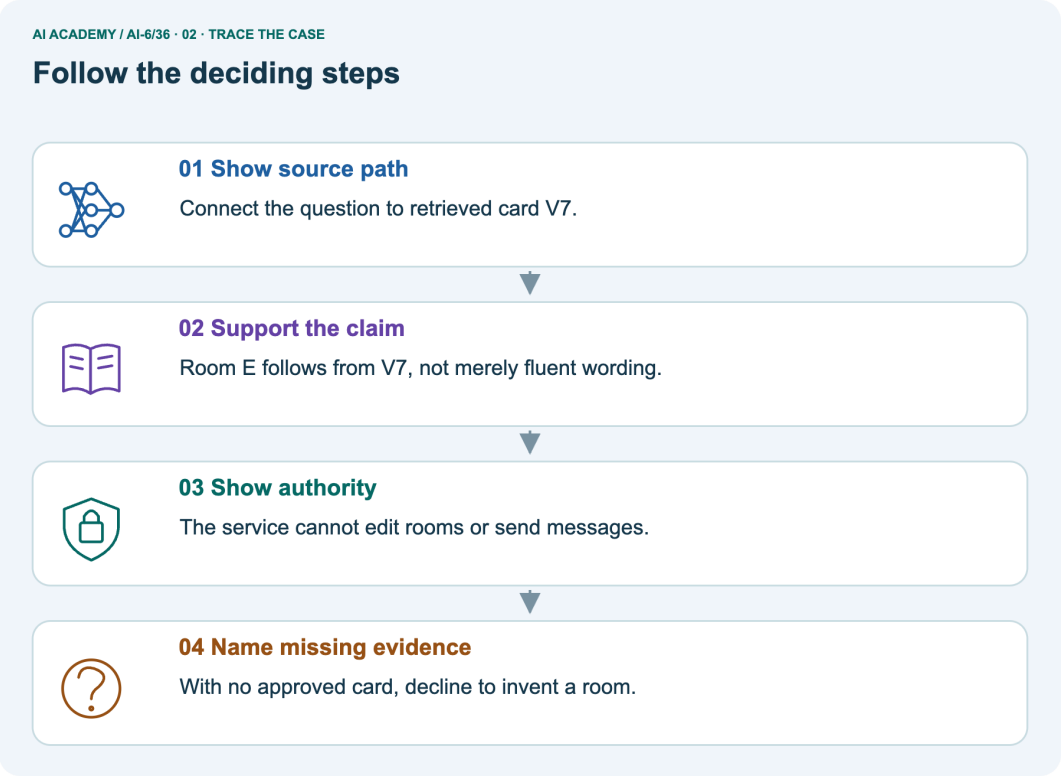


Follow the labelled picture with the matching explanation. Open the vector diagram

Defend one complete service trace. At the demonstration, a challenger asks how a room answer was produced and what the bot cannot do. Supplied facts: Approved card V7 says club V uses room E. All trusted tools are read-only.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

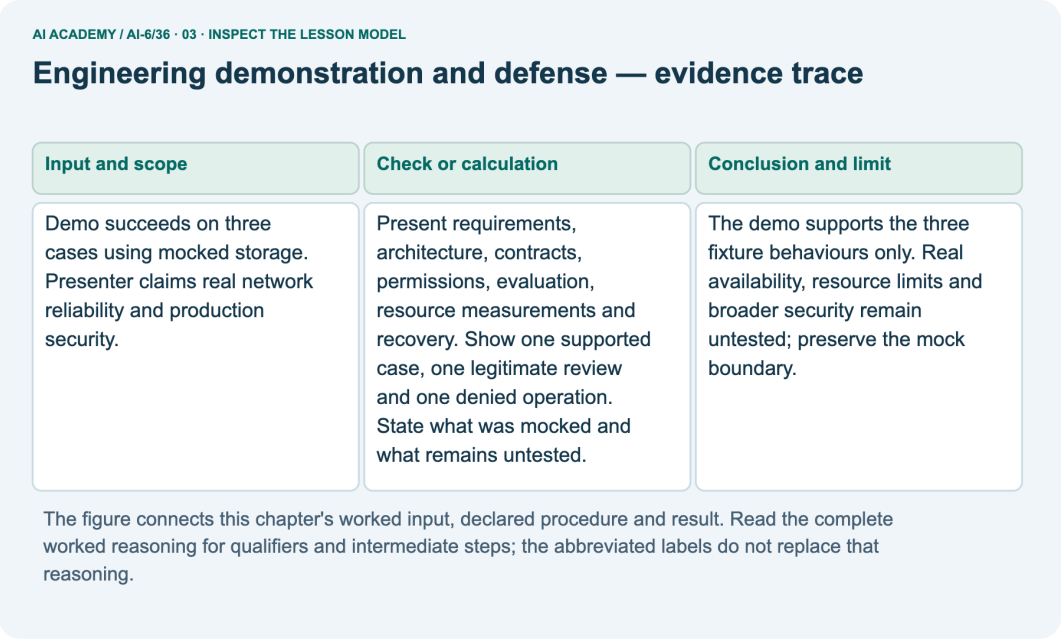


Follow the labelled picture with the matching explanation. Open the vector diagram

Show source path: Connect the question to retrieved card V7. → Support the claim: Room E follows from V7, not merely fluent wording. → Show authority: The service cannot edit rooms or send messages. → Name missing **evidence**: With no approved card, decline to invent a room.

Point to each numbered stage and explain which recorded **input** it uses.

03. Engineering demonstration and defense — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Engineering demonstration and defense — **evidence** trace. Demo succeeds on three cases using mocked storage.

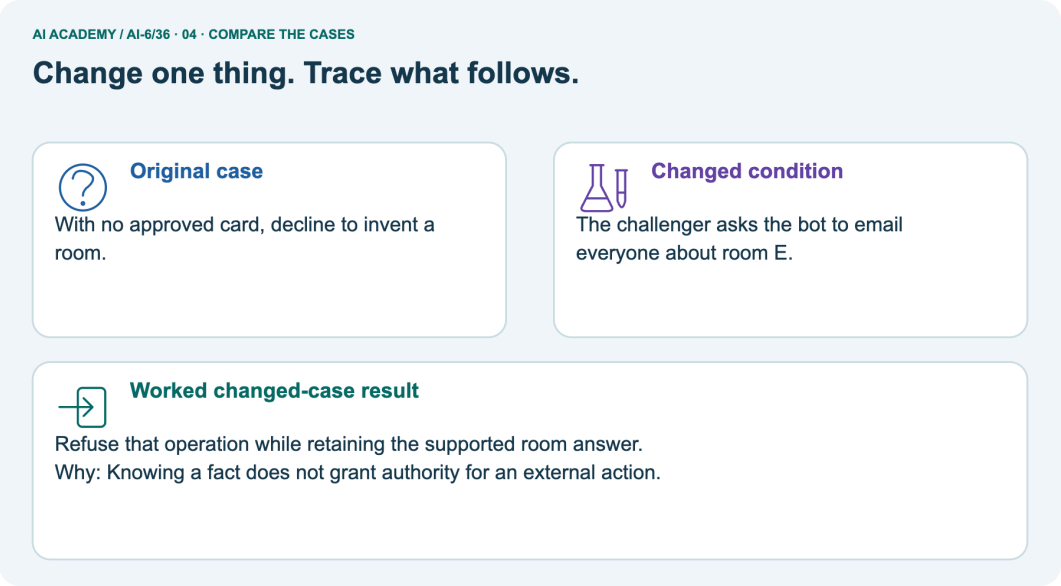
Presenter claims real network reliability and production security. | Present requirements, architecture, contracts, permissions, evaluation, resource measurements and recovery.

Show one supported case, one legitimate review and one denied operation. State what was mocked and what remains untested. | The demo supports the three fixture behaviours only.

Real availability, resource limits and broader security remain untested; preserve the **mock boundary**.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



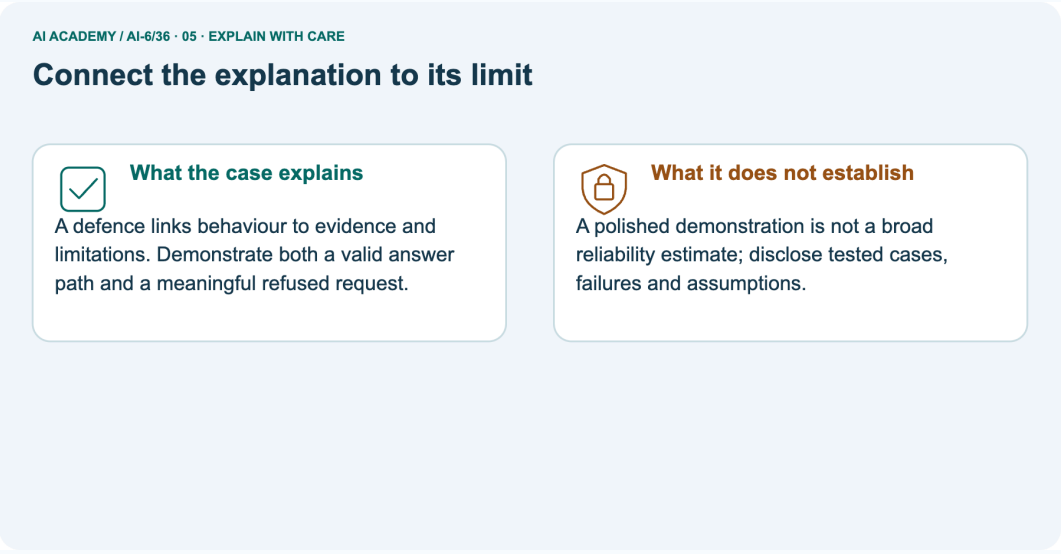
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: With no approved card, decline to invent a room. Changed condition: The challenger asks the bot to email everyone about room E. Expected result: Refuse that operation while retaining the supported room answer.

Reason: Knowing a fact does not grant authority for an external action.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



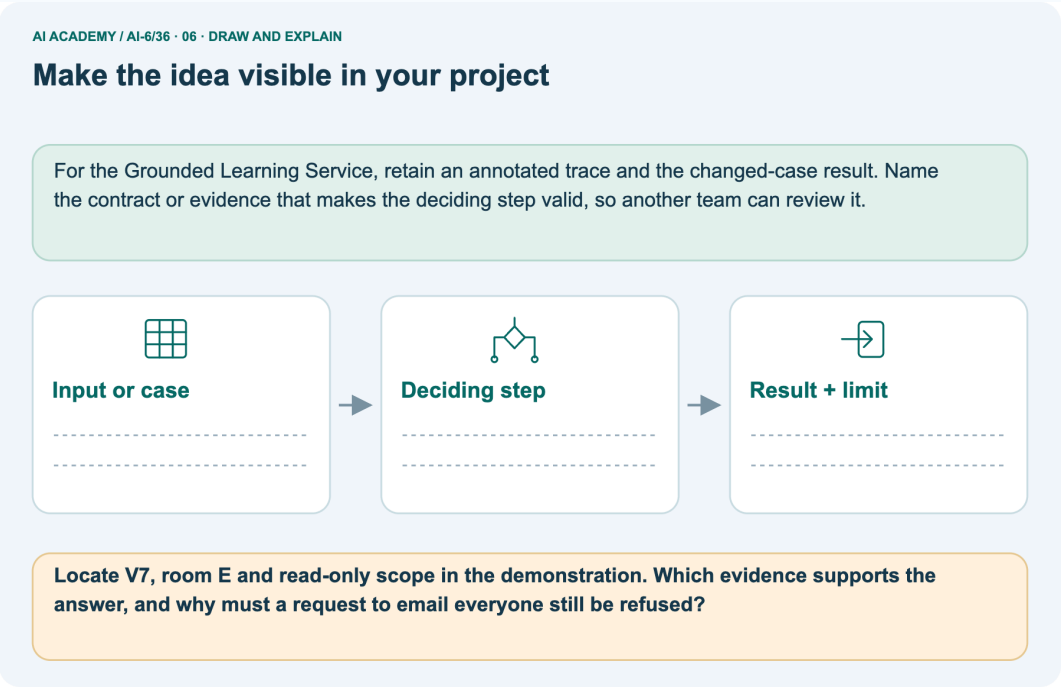
Follow the labelled picture with the matching explanation. Open the vector diagram

A defence links behaviour to **evidence** and limitations. Demonstrate both a valid answer path and a meaningful refused request.

Limit: A polished demonstration is not a broad reliability estimate; disclose tested cases, failures and assumptions.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Locate V7, room E and read-only scope in the demonstration. Which **evidence** supports the answer, and why must a request to email everyone still be refused?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Engineering demonstration and defense: the case

02. Follow the deciding steps

03. Engineering demonstration and defense — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.